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Navigating the Depths in Offshore Andaman, Indonesia: Revolutionizing Deepwater Drilling with Seismic While Drilling and Fiber-Optic Technologies

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Abstract

This paper presents a comprehensive technical evaluation of Mubadala Energy's deepwater exploration campaign in the Offshore Andaman Basin, Indonesia, where an integrated suite of Logging While Drilling (LWD), Seismic While Drilling (SWD), and fiber-optic Distributed Acoustic Sensing (DAS) technologies was deployed to address the multifaceted challenges of frontier drilling. The wells, drilled between 2023 and 2024 in water depths of 1,200 meters, targeted the Oligocene Bampo reservoir in one of Southeast Asia's most geologically complex deepwater provinces.

The primary objective was to mitigate key subsurface uncertainties, including pore pressure prediction in overpressured shales, reservoir depth control in areas with ± 100 meters seismic uncertainty, and efficient seismic acquisition under remote offshore logistics. The integration of Quadrupole Sonic, Resistivity at the Bit (RAB), and SWD enabled real-time formation evaluation and pressure monitoring, while DAS technology allowed for wireline-based vertical seismic profiling with significantly reduced operational footprint.

Real-time SWD look-ahead imaging reduced reservoir depth uncertainty to ± 10 meters, eliminating the need for an 11%-inch contingency liner—resulting in **USD 8.4 million** in direct cost savings. Optimized DAS deployment cut borehole seismic acquisition time by over 50%, while improving seismic resolution and reducing environmental impact through lower rig time and helicopter usage.

This case study demonstrates how an integrated LWD-SWD-DAS approach not only improved safety, efficiency, and geological accuracy, but also enabled proactive, data-driven decision-making in a high-risk exploration setting. The results establish a new benchmark for deepwater exploration drilling in Indonesia and provide a transferable framework for other frontier basins globally.

Introduction

The global energy demand continues to push the boundaries of oil and gas exploration into increasingly complex and remote environments. Among the most technically challenging and operationally risky are deepwater frontiers, where geological uncertainty, high pressure regimes, and narrow drilling margins place a premium on efficiency, safety, and precision. Deepwater Indonesia, specifically the Andaman Sea off the

northwest coast of Aceh Province, represents one such frontier. It offers significant hydrocarbon potential but is characterized by structurally complex geology, limited well control history, and a paucity of offset well data to guide operations.

To address these challenges, Mubadala Energy drilled three vertical exploration wells in the offshore Andaman field between 2022 and 2024. These wells, drilled in water depths exceeding 1000 meters, targeted the Oligocene Bampo 26Ma clastic reservoir. Subsurface uncertainty was especially pronounced, with the predicted reservoir top depth carrying an uncertainty margin of ± 100 meters True Vertical Depth (TVD). Such uncertainty posed a substantial risk of penetrating the reservoir unintentionally, potentially leading to kicks or well control incidents. Therefore, precise, real-time subsurface data acquisition was not just beneficial but essential. Mubadala Energy responded to this challenge by adopting a technology-forward approach that integrated advanced Logging While Drilling (LWD) and borehole seismic techniques as the foundation for safe and informed drilling.

Traditionally, drilling operations relied on pre-drill seismic models and offset well data for geosteering and casing point selection. However, these models often lack the fidelity needed to manage dynamic, high-risk drilling environments. Recognizing the limitations of conventional workflows, the team shifted from a reactive to a proactive drilling strategy—one that emphasized real-time formation evaluation and subsurface imaging to support faster, safer decision-making.

At the core of this strategy was the deployment of a comprehensive suite of advanced LWD technologies. A Quadrupole Sonic (QST) tool was used for geomechanical analysis and pore pressure prediction; a Seismic While Drilling (SWD) tool provided look-ahead imaging and enabled real-time time-depth calibration; and a Resistivity at the Bit (RAB) tool delivered granular formation evaluation to optimize coring point selection. This suite of tools allowed the drilling team to constantly update the geological model with real-time data from the bottom-hole assembly (BHA), significantly reducing uncertainties during critical operations such as setting casing or drilling near anticipated pressure transitions.

The campaign also marked a world-first deployment of dual-mode seismic acquisition in this type of environment—merging SWD and wireline (WL) fiber-optic seismic technology using Distributed Acoustic Sensing (DAS). During drilling, checkshot data was acquired in real time via the SWD tool, allowing for dynamic calibration of the velocity model. Following Total Depth (TD), a fiber-optic heptacable was deployed during planned wireline logging runs, transforming the entire wellbore into a continuous seismic sensor. This method enabled the acquisition of full-depth seismic profiles in a single pass, eliminating the need for station-by-station measurements and greatly enhancing operational efficiency.

These technological innovations were not theoretical; they delivered measurable improvements in operational outcomes. The SWD tool helped the team visualize formation tops while drilling, ensuring the 9-5/8-in casing could be set as close as safely possible to the reservoir top—critical for maximizing kick tolerance and maintaining robust well control. Real-time integration of seismic and resistivity data enabled the team to adapt the drilling strategy on the fly, avoid potential hazards, and optimize formation evaluation programs—all without introducing additional non-productive time (NPT).

This paper presents a detailed account of the methodology, execution, and outcomes from this deepwater drilling campaign. It highlights how a technology-driven approach—anchored in advanced LWD, SWD, and fiber-optic systems—can reduce subsurface uncertainty, enhance operational safety, and unlock greater efficiency in deepwater exploration. The results offer a roadmap for future campaigns in geologically complex and high-stakes environments, where conventional approaches often fall short.

Objectives

The primary objective of this deepwater exploration campaign was to drill safely and efficiently in a high-risk, geologically complex environment. The application of an integrated advanced Logging While Drilling

(LWD) and borehole seismic suite was central to achieving this goal. The following specific objectives guided the planning and execution of the campaign:

- **Enhance Real-Time Formation Evaluation and Decision-Making**

Move beyond reliance on static pre-drill models by acquiring dynamic, real-time data on lithology, porosity, and fluid content. This allowed geologists and petrophysicists to make immediate subsurface interpretations and adjust drilling parameters or casing plans accordingly, minimizing the risk of unexpected formation encounters.

- **Reduce Operational Risks Through Accurate Pore Pressure and Formation Evaluation**

Address the challenge of narrow drilling margins by continuously updating the pore pressure model using high-resolution LWD sonic and seismic data. This enabled precise mud weight management to mitigate the risks of wellbore instability, kicks, and losses. In particular, real-time data was used to inform decisions regarding whether to trigger a pre-defined contingency plan involving an additional liner and hole enlargement—an outcome with significant time and cost implications.

- **Optimize Casing Setting Depth Using Look-Ahead Seismic**

Given the significant uncertainty in the depth of the reservoir top (± 100 m TVD), one of the most critical objectives was to set the 9- 5/8-in casing as close as safely possible above the reservoir. Seismic While Drilling (SWD) data enabled real-time visibility of formation tops ahead of the bit, reducing the risk of penetrating the reservoir without adequate well control barriers and enhancing overall kick tolerance.

- **Improve Reservoir Characterization and Core Point Selection**

Utilize real-time Resistivity at the Bit (RAB) measurements to identify geologically representative intervals for coring. This helped optimize coring operations, reducing wasted rig time and ensuring high-quality reservoir samples were collected for post-well analysis.

- **Demonstrate the Operational Benefits of Dual-Mode Seismic Acquisition**

Deploy fiber-optic seismic technology via a heptacable during planned wireline logging runs to collect full-wellbore seismic data using Distributed Acoustic Sensing (DAS). By comparing this with SWD-acquired data, the campaign aimed to evaluate the feasibility and efficiency of this method for future deepwater operations. The fiber-optic system enabled seismic acquisition across the entire borehole in a single pass, eliminating time-consuming station-by-station wireline checkshots and significantly improving seismic coverage and operational efficiency.

Each of these objectives was aimed at minimizing uncertainty, avoiding unplanned events, and improving operational performance. Collectively, they demonstrate how advanced LWD and seismic technologies, when integrated into a cohesive drilling strategy, can significantly enhance safety, efficiency, and reservoir understanding in deepwater exploration environments.

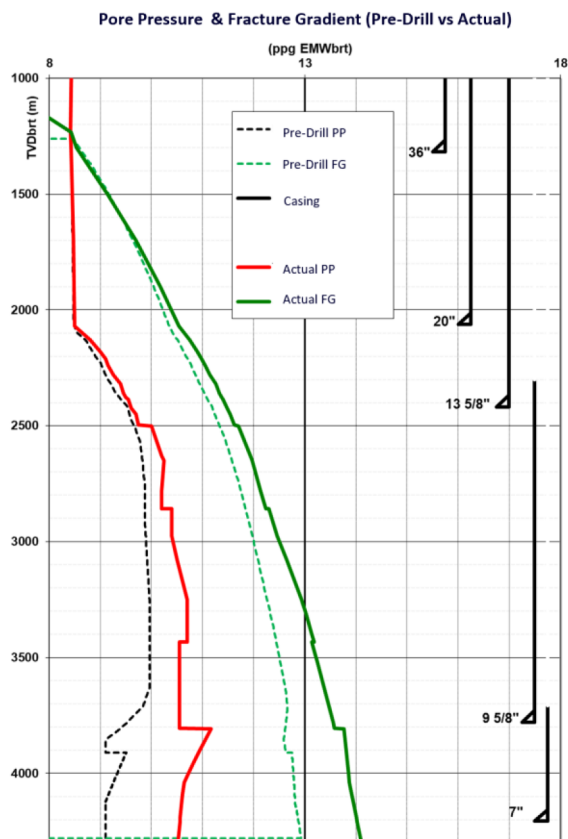


Figure 1—Comparison of pre-drill and updated real-time PP-FG models. The integration of Quadrupole Sonic and Seismic While Drilling data enabled dynamic calibration of the pore pressure prognosis while drilling. This real-time insight allowed proactive mud weight adjustments and optimization of casing depth, significantly reducing risk of unexpected pressure ramp and enhancing operational decision-making.

Methodology

The Andaman deepwater drilling campaign leveraged an integrated approach that combined advanced Logging While Drilling (LWD), real-time formation evaluation, and dual-mode seismic acquisition technologies. The methodology was designed to systematically reduce geological uncertainty, enhance operational safety, and support real-time decision-making in a frontier deepwater setting.

Well Design and Drilling Strategy

Each well was designed with a conservative casing plan to allow for contingencies driven by subsurface uncertainty. In particular, the plan incorporated the potential need to enlarge the 12-1/4-in hole section to 14-3/4-in and run an 11-3/4-in liner if real-time formation evaluation indicated high pore pressure or instability risks. To inform such decisions in real time, the well design integrated a suite of advanced downhole tools deployed during the 12-1/4-in and 8-1/2-in hole sections.

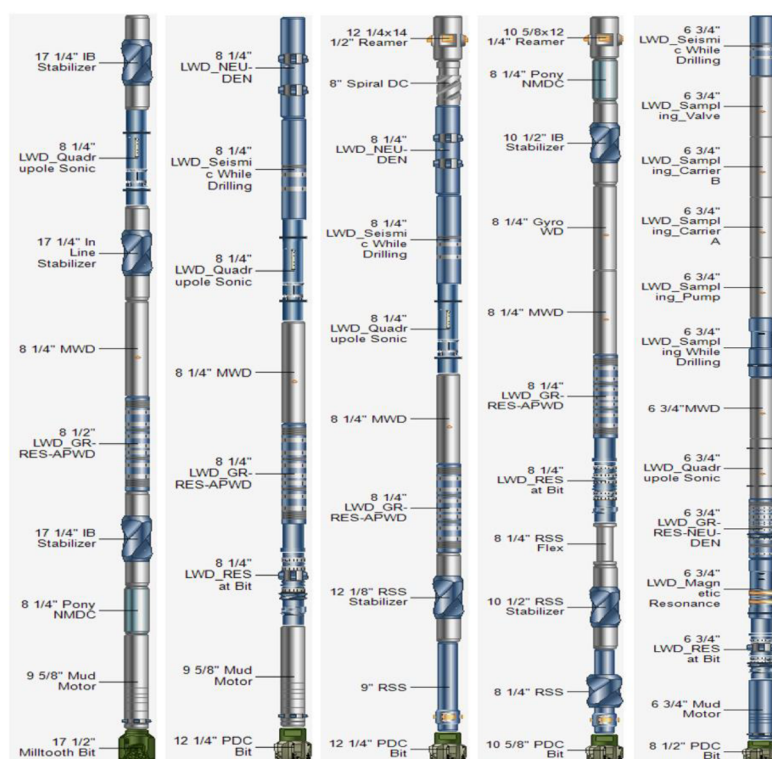


Figure 2—BHA configurations for deepwater wells showcasing the integration of advanced Logging While Drilling (LWD) tools including Quadrupole Sonic, Seismic While Drilling, Resistivity at Bit, and Formation Sampling. Each BHA was tailored to specific geological objectives, balancing real-time formation evaluation and drilling performance.

Logging While Drilling (LWD) Deployment

Three critical LWD technologies were deployed:

Quadrupole Sonic Tool (QST): The Quadrupole Sonic Tool (QST) served as a cornerstone of the real-time formation evaluation and geomechanical analysis program during drilling. Its capability to record compressional, shear, and Stoneley wave velocities enabled accurate determination of critical rock properties such as porosity, permeability indicators, and mechanical strength.

The QST operates through both monopole and quadrupole acoustic sources. Monopole transmitters produce compressional and Stoneley waves—useful for porosity and fracture/permeability assessment—while quadrupole transmitters generate flexural waves particularly effective in identifying shear velocities, even in soft or unconsolidated formations typical of deepwater environments. The tool's 12 axial receiver stations and 48 azimuthally spaced sensors allow for accurate wavefield decomposition and anisotropy detection.

A major application of the QST in this campaign was **real-time pore pressure prediction**. By continuously measuring Delta-T (compressional slowness), deviations from the expected Normal Compaction Trend (NCT) were monitored to detect overpressured zones. These real-time insights allowed proactive mud weight adjustments to stay within the safe mud weight window, avoiding kicks or losses. The accuracy of lithological boundaries derived from QST also enabled better calibration of the NCT throughout the well. This data was foundational for wellbore stability modeling and for minimizing risk in high pressure high temperature (HPHT) sections.

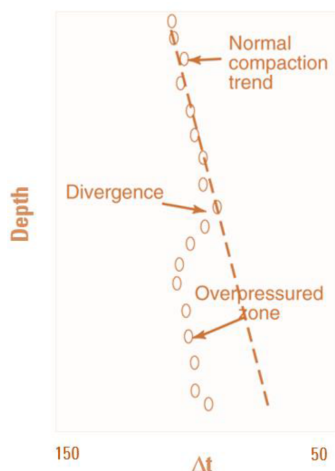


Figure 3—Deviation from the Normal Compaction Trend marks the onset of overpressure, enabling real-time pore pressure prediction ahead of the bit.

Seismic While Drilling (SWD): the **Seismic While Drilling (SWD)** technology provided an invaluable real-time look-ahead capability by acquiring seismic checkshot and vertical seismic profile (VSP) data during drilling. This enabled calibration of the velocity model and significantly reduced uncertainty in formation depth prediction—especially critical when drilling exploration wells in frontier areas where pre-drill seismic data may have vertical uncertainties of 100-150 meters.

The SWD system works by detecting seismic pulses emitted from a surface airgun and received by a downhole sensor integrated in the BHA. As drilling progresses, checkshots and seismic reflection data are captured in real time and used to perform well ties and look-ahead modeling. This real-time imaging was especially critical when approaching the primary reservoir top, where precise depth control was required to optimally place the 9-5/8 in. casing. By providing a look-ahead of ~100 meters, SWD significantly reduced the need for conservative casing depths and minimized the drilled contingency footage—leading to time and cost savings.

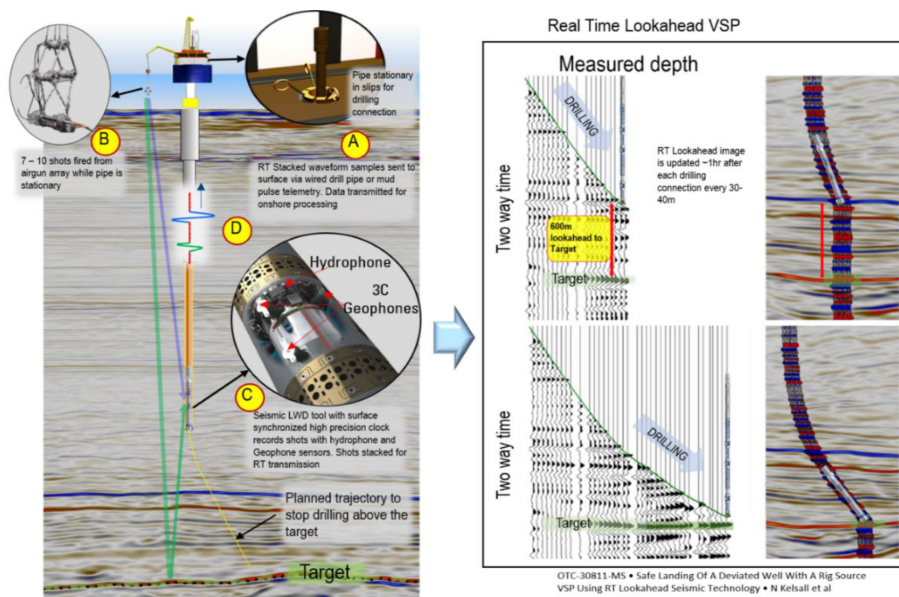


Figure 4—SWD Real Time Acquisition Sequence (Kelsall et al., 2020)

The SWD process included:

- Pre-survey modeling based on pre-drill data and formation tops.
- Real-time checkshot acquisition at each stand.
- Iterative corridor stack imaging to detect approaching reflectors.
- Post-run processing for VSP QC and seismic well tie.

Resistivity at the Bit (RAB): The **RAB tool** delivers continuous formation resistivity measurements directly at the drill bit, enabling early formation evaluation even before the BHA clears the section. This is particularly advantageous in time-critical environments where real-time data informs drilling decisions such as coring, casing point selection, and wellbore trajectory adjustments.

The tool functions by driving a low-frequency axial current into the formation, with the bit acting as part of the measure electrode. This creates an unfocused measurement of resistivity in the immediate vicinity of the bit, allowing identification of lithological boundaries, fluid contacts, and formation tops as they are drilled.

In this project, RAB data enabled:

- Differentiation between hydrocarbon- and water-bearing zones based on resistivity contrast.
- Identification of optimal coring points by pinpointing reservoir intervals in real-time.
- Geostopping decisions to optimize casing setting depths while minimizing overdrilling of key reservoir units.

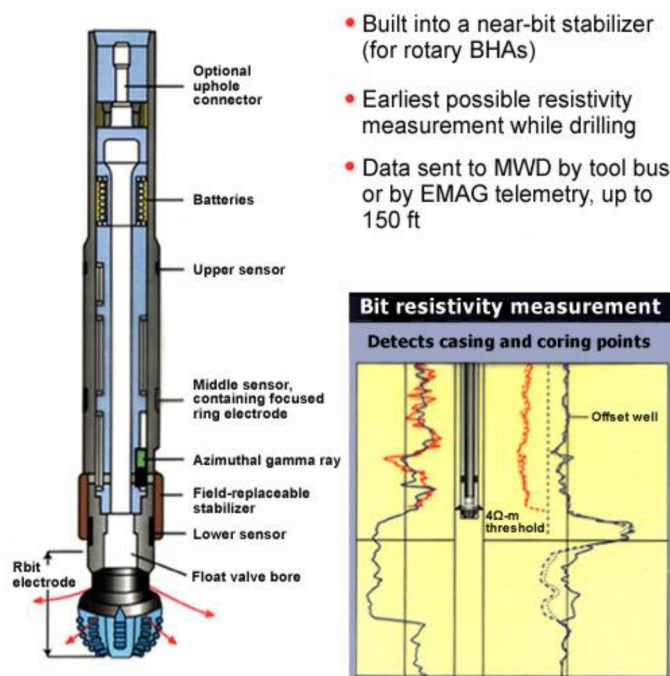


Figure 5—Near-bit resistivity tool design enabling the earliest possible resistivity measurement while drilling, improving formation evaluation and early detection of casing/coring points.

Real-Time Data Integration and Decision-Making

A centralized Operations Support Center (OSC) team received LWD and SWD data in real time. The team, comprising geologists, geophysicists, and drilling engineers, used the data to:

- Continuously refine the pore pressure model based on sonic velocities.
- Adjust mud weights and casing setting depths in response to live geomechanical indicators.
- Evaluate the need for activating the casing contingency plan.
- Calibrate the time-depth relationship using SWD look-ahead travel times and update the anticipated reservoir top depth.

Key decisions, such as setting the 9-5/8-in casing just above the reservoir, were made based on the integrated interpretation of these data streams.

Wireline Seismic and Fiber-Optic Deployment

Following the drilling of the 8-1/2-in reservoir section to Total Depth (TD), wireline logging operations included the deployment of a heptacable equipped with optical fibers to enable Distributed Acoustic Sensing (DAS). This fiber-optic-enabled cable effectively transformed the entire wellbore into a continuous seismic sensor, allowing for a novel method of vertical seismic profiling (VSP).

A surface airgun array was utilized to generate seismic energy, which was recorded along the entire fiber-optic-enabled cable using the DAS system. Unlike traditional wireline VSPs that acquire seismic data at discrete depths using a downhole geophone, DAS enabled full-depth seismic acquisition in a single pass, significantly enhancing operational efficiency and data resolution.

Fiber-Optic DAS represents a transformative approach to borehole seismic acquisition. Instead of relying on conventional geophone tools, DAS systems use Rayleigh backscattering of laser pulses transmitted through the optical fiber. Strain-induced phase shifts in the backscattered light are measured and time-depth correlated, effectively converting the fiber into an array of densely spaced vibration sensors.

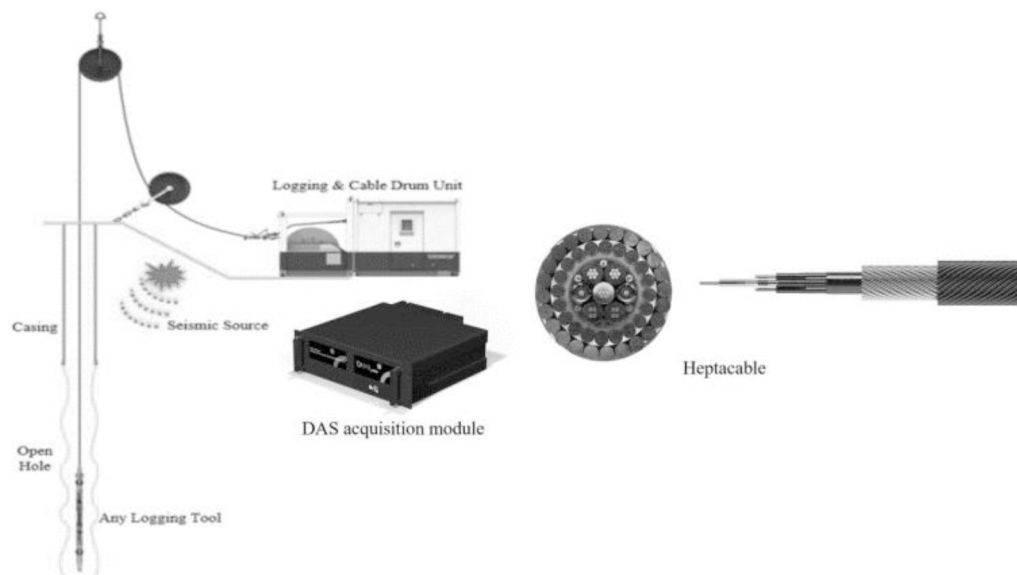


Figure 6—Fiber Optic Deployment

The deployment of DAS in a deepwater environment offered several key benefits:

- **Operational Efficiency:** Eliminated the need for multiple depth-stationed geophone recordings, saving rig time and reducing risk exposure.
- **Full-Wellbore Seismic Imaging:** Enabled continuous seismic sensing along the wellbore, improving vertical resolution and providing valuable data for tie-in with surface seismic.

- **Data Stacking Capability:** Allowed for repeated airgun shots to be stacked while the fiber remained stationary, enhancing signal-to-noise ratio and data fidelity.
- **Improved Safety and Cost Control:** Provided a cost-effective solution for acquiring VSP data in high-risk, deepwater settings, especially valuable in exploration wells with limited seismic calibration.

By integrating DAS-acquired seismic data with the real-time SWD data, the team was able to cross-validate time-depth relationships and enhance confidence in subsurface interpretation. This dual-mode seismic acquisition strategy marked a first-of-its-kind deployment in the region and demonstrated the viability of fiber-optic-enabled wellbore sensing in offshore exploration.

Cross-Validation of Seismic Technologies

To evaluate the effectiveness and reliability of the novel fiber-optic DAS system, seismic data acquired using the wireline DAS were directly compared to data acquired using SWD. This cross-validation ensured confidence in the results and provided insights into the strengths and limitations of each approach.

The deployment of this integrated toolset was not only about acquiring data—it was about enabling better decisions. By shortening the feedback loop between measurement and action, the campaign exemplified how advanced LWD and SWD technologies can transform subsurface uncertainty into real-time operational intelligence.

Results

The integrated application of Quadrupole Sonic Tool (QST), Seismic While Drilling (SWD), Resistivity at the Bit (RAB), and post-TD fiber-optic-enabled wireline seismic in this deepwater exploration campaign delivered exceptional results across safety, operational efficiency, and reservoir evaluation. These outcomes directly addressed the pre-drill challenges and demonstrated the tangible value of real-time, data-driven operations.

Pore Pressure and Liner Optimization

The primary pre-drill risk in the 17-1/2 in. and 12-1/4 in. hole sections was the significant uncertainty in the pore pressure profile. In conventional deepwater practice, operators often include one or more contingency liner sections in the well design to manage potential pressure ramps. However, each additional liner adds considerable time and cost to the operation. In this campaign, a contingency 11-3/4 in. liner was prepared for potential deployment between the 12-1/4 in. x 14-1/2 in. and 10-5/8 in. x 12-1/4 in. hole sections.

To address this risk, real-time compressional slowness data from the QST—combined with checkshot velocities from the SWD—enabled continuous and accurate calibration of the pore pressure model. The workflow followed a three-step process:

1. **Real-Time Monitoring:** Observed AT (compressional slowness) from the QST was plotted against a pre-defined Normal Compaction Trend (NCT) for the basin.
2. **Deviation Analysis:** Departures from the NCT were identified in real-time and quantified using the Eaton method, yielding a dynamic pore pressure prediction ahead of the bit.
3. **Model Calibration:** SWD checkshots provided independent velocity measurements to anchor the time-depth relationship and refine the overburden stress calculations.

The real-time pore pressure monitoring revealed that the pre-drill model was conservative. The entire pore pressure and fracture gradient (PP-FG) window was shifted approximately 0.5 ppg to the right relative to the prognosis. This meant we encountered both higher pore pressure and a higher fracture gradient than anticipated.

In this exploration context, the 11%-inch contingency liner was designed as a mitigation for multiple risks: not only a severely narrow PP-FG window but also the potential for losses in stringer formations or the need to isolate an unstable section if the 13-5/8 in. casing shoe placement was suboptimal.

Critically, the integration of real-time QSonar and SWD data showed that while the entire window had shifted to higher pressures, the drilling margin between PP and FG remained stable and workable. The data confirmed the absence of the other contingency triggers: no problematic stringers were encountered ahead of the bit, and the formation strength at the 13-5/8 in. casing shoe was ideal. This holistic, real-time dataset provided the quantitative confidence to proactively de-scope the contingency liner from the well program. The resultant savings of USD 8.4 million are attributed to this data-driven decision, which eliminated the need for the liner, its associated hole enlargement run, and the related rig time. In similar frontier exploration wells without this integrated real-time capability, running such a contingency liner is a standard and costly risk-mitigation practice.

This approach enabled data-driven decisions on whether to drill contingency hole sections or proceed with liner runs. As a result, the well construction process was streamlined, saving up to seven days of rig time. At a rig spread rate of USD 1.2 million per day, this translated into direct cost savings of over USD 8.4 million. Furthermore, the continuous PPFG monitoring—based on real-time compressional and shear slowness data—enabled optimal mud weight and ECD selection, effectively mitigating risks of wellbore instability, fluid losses, and well control events.

Seismic Lookahead and GeoStopping

SWD technology provided a game-changing capability: real-time seismic lookahead while drilling. Once at least ten waveform stations spanning 700 meters were acquired at drilling connections, SWD VSP processing could begin. The system generated continuous updates of one-way seismic travel times between the tool and upcoming formation reflectors. By converting seismic time to depth using pre-drill velocity models constrained by measured behind-the-bit velocities, the uncertainty in the reservoir top depth was reduced from ± 100 -150 meters to as little as ± 15 meters when the tool was within ~ 120 meters above the target. This was enabled by iterative time- to-depth conversion techniques, guided by SWD-acquired velocities and refined using trough-to-trough and break-to-break time picking methods. A decision-tree workflow helped select the best approach, minimizing prediction error.

The value of this real-time imaging was immediately quantifiable. The pre-drill model placed the target reservoir top significantly deeper than actual, with an uncertainty of ± 100 m. This led to a conservatively planned 9-5/8 in. casing point set intentionally high. Real-time SWD imaging, initiated approximately 150m above this prognosis, indicated the target was over 100 meters shallower than expected. This allowed the team to confidently drill an additional 10m of section and optimally set the casing 30m closer to the reservoir—just ~ 25 m above the refined SWD prediction.

This enhanced accuracy enabled precise casing point selection just above the reservoir. It significantly reduced the risk of unintentional reservoir penetration and subsequent well control challenges, while also eliminating the need for excessive contingency footage. The approach also avoided traditional time-drilling and delayed cuttings analysis, thus saving additional rig time.

These capabilities were further validated using Recorded Mode (RM) SWD data in the 8-1/2 in. hole section, which correlated well with fiber-optic seismic results. The subsequent DAS-VSP acquisition ("Optiq seismic"), considered the definitive measurement, confirmed the reservoir top. The SWD prediction was therefore within 2 meters of the ground truth, a remarkable validation of its accuracy. Event alignment between datasets confirmed the reflector position, while velocity profiles from SWD, DAS, and sonic logs demonstrated high consistency, increasing operator confidence in formation depth estimates. Additionally, the SWD processing turnaround—from waveform acquisition to interpreted reflector depth—was achieved within 6 hours of data capture. This enabled same-day decisions by the on-site drilling team, supported remotely by geophysicists via real-time WITSML data feed. The collaborative interpretation workflow

significantly compressed the decision-making cycle, improving operational agility in a high-stakes offshore environment.

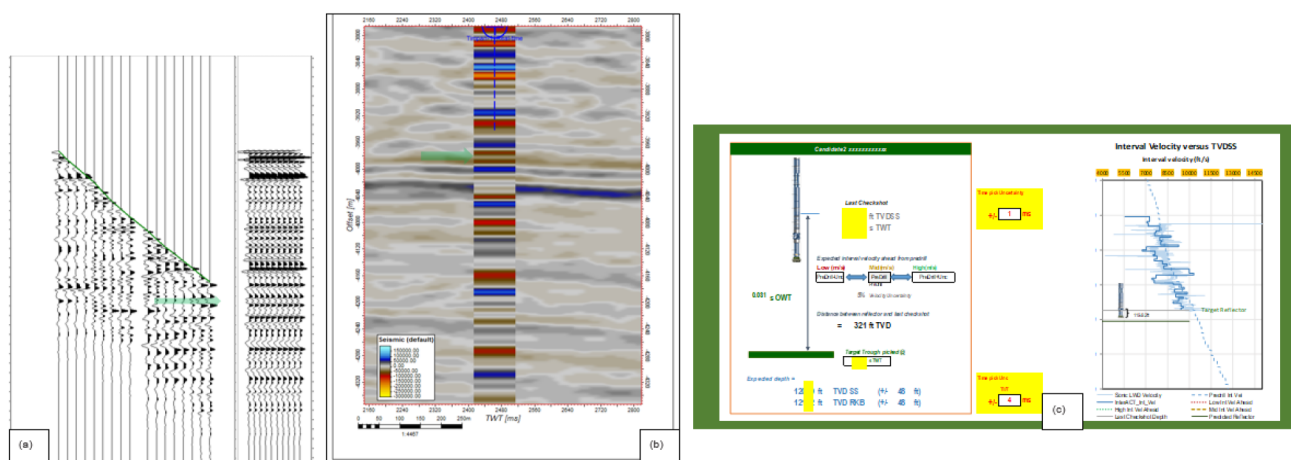


Figure 7—Example of RT upgoing compressional wavefield with highlighting of formation target, along with seismic well tie (middle) and lookahead prediction (left) (Sakina et al., 2024)

Core Sampling and Reservoir Evaluation

Capturing high-quality reservoir core in deepwater exploration is particularly challenging due to the tight and unpredictable nature of the formations. Conventional methods often result in missed targets or low recovery due to reliance on low-resolution or pre-drill models.

In this campaign, the RAB tool's resistivity-at-the-bit measurements proved transformative. Unlike other tools higher in the BHA, RAB provided real-time resistivity readings at the exact point of bit penetration. Sudden changes in resistivity signaled formation entry, enabling precise identification of the top of the reservoir.

This allowed the drilling team to initiate coring operations immediately upon entering the target zone, ensuring the core was extracted from the most valuable and representative part of the reservoir. The result was a high-quality core sample that significantly improved post-well reservoir analysis and reduced uncertainties in rock properties, saturation, and depositional environment.

This real-time coring strategy addressed a common weakness in offshore coring—poor correlation between coring intervals and actual reservoir properties—thus delivering higher value insights for future development planning.

Wireline Seismic and Fiber-Optic Deployment

Following TD in the 8-1/2 in. reservoir section, a wireline logging run was conducted using a hybrid hepta-cable incorporating fiber optics. This enabled Distributed Acoustic Sensing (DAS), transforming the wellbore into a continuous seismic sensor.

A surface airgun array was deployed to generate seismic waves, and the DAS system recorded the acoustic response along the wellbore in real-time. Unlike conventional wireline vertical seismic profiles (VSPs), which require repeated tool repositioning, the DAS system acquired a full-depth VSP in a single stationary run, greatly enhancing both efficiency and seismic resolution.

To ensure optimal coupling of the cable with the casing, a slack cable test was performed at different depths (30 m, 60 m, and 80 m), revealing that slack application significantly improved signal-to-noise ratio. However, improper slack could lead to coupling imperfections and nonreal reflections, which were mitigated through depth correction algorithms prior to processing.

The raw fiber-optic data underwent a standard zero-offset VSP processing workflow, including wavefield separation and deconvolution. This produced a high-quality upgoing P-wavefield, which clearly identified

the target reflector. The final corridor stack, created from this signal, showed excellent agreement with the synthetic seismogram generated from calibrated sonic and density logs. This confirmed the seismic well-tie and validated the slack correction applied.

The DAS-acquired data also displayed a frequency content range of 5-80 Hz, higher than typical conventional seismic, allowing improved vertical resolution. When compared with SWD datasets and sonic logs, the fiber-optic velocity profile closely matched, reaffirming the structural interpretation and depth accuracy.

Integration and Impact

The benefit of combining SWD and fiber-optic seismic was the ability to cross-validate datasets in real-time and post-drill. As shown in travel time and velocity profile comparisons, the average discrepancy between SWD and fiber-optic TWT measurements was only ~4 ms at similar depths. Upgoing P-wavefield events aligned well across methods, including the identified target reflector.

Collectively, the integration of these advanced LWD and post-drill technologies enabled real-time, high-resolution insight into the subsurface. This empowered the drilling team to reduce operational risks, improve efficiency, and maximize reservoir understanding—delivering a successful campaign outcome in one of the most challenging offshore basins in Southeast Asia.

Conclusion

The successful deployment of an integrated suite of Logging While Drilling (LWD) technologies—including Quadrupole Sonic (QST), Seismic While Drilling (SWD), Resistivity at the Bit (RAB), and fiber-optic-enabled Distributed Acoustic Sensing (DAS)—marked a transformative leap in the way deepwater exploration wells are drilled and evaluated in Indonesia. This campaign exemplified a proactive, real-time, and data-driven drilling approach that fundamentally reshaped subsurface risk management, well construction decisions, and reservoir evaluation.

One of the most impactful outcomes was the ability to dynamically refine pore pressure and fracture gradient models using real-time QST data. This allowed the operator to safely eliminate a contingency casing string, translating into a rig time savings of approximately **7 days**, and a direct cost saving of **USD 8.4 million**, based on a spread rate of USD 1.2 million per day. Such proactive decision-making would not have been possible with conventional drilling practices relying solely on offset models and lagging wireline data.

SWD technology played a critical role in de-risking the well's final casing point and ensuring optimal placement just above the target reservoir. The depth uncertainty to the primary reservoir was reduced from ± 100 meters to as little as ± 15 meters. This capability enabled safer, more efficient drilling across the overpressured intervals and allowed the team to maximize reservoir exposure. Further seismic refinement was made possible through a novel application of DAS using a fiber-optic-enabled wireline heptacable. This allowed for full-borehole vertical seismic profile (VSP) acquisition in a single run, without the need for dedicated downhole seismic tools. Not only did this approach save **~15 rig hours per well**, but it also contributed to an estimated **emission reduction of ~30.27 tonnes of CO₂ per well**, aligning with the operator's decarbonization goals.

In the reservoir section, the RAB tool enabled the selection of optimal coring points, ensuring recovery of the most representative formation samples. This real-time resistivity mapping proved invaluable for high-grading intervals for reservoir modeling, helping to extract maximum geological value from a single core. The integration of physical core data with continuous LWD logs provided a richer understanding of lithology, porosity, and reservoir connectivity.

Throughout the three-well campaign, workflows were iteratively improved from one well to the next based on real-time learnings. This operational agility—made possible by robust communication between rig site personnel, subsurface experts, and remote support teams—was key to maximizing the benefit of

each technology. Lessons learned were quickly embedded into subsequent wells, optimizing both tool deployment and decision-making processes.

This study illustrates that LWD technologies are no longer just diagnostic tools—they are operational enablers that redefine how exploration wells are drilled. The synergistic integration of sonic, seismic, resistivity, and fiber-optic data created a comprehensive, high-resolution view of the subsurface, empowering drilling and subsurface teams to act decisively under pressure.

Ultimately, this campaign set a new benchmark for safe, efficient, and intelligent deepwater drilling in Southeast Asia. It highlights that strategic investment in integrated LWD and borehole seismic solutions is not only cost-effective but also indispensable for tackling the geological and operational challenges of frontier deepwater plays. The experience and outcomes of this campaign provide a robust framework for future wells in Indonesia and other complex offshore basins around the world.

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